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Engineering Mathematics

11Week [13 November - 19 November]

Quiz 11

Started on	2023-11-20 08:26
State	Finished
Completed on	2023-11-20 08:28
Time taken	1 min 50 secs
Grade	4.00 out of 5.00 (80%)

Question 1

Correct

Mark 1.00 out of 1.00

If we eliminate y from the system

$$\begin{cases} x' + y = 0, & x(0) = \frac{7}{4} \\ 4x + y' = 3, & y(0) = 4 \end{cases}$$

then we obtain the following IVP:

Select one:

- ☐ $x'' - 4x = 3, \quad x(0) = \frac{7}{4}, \quad x'(0) = 4$
- ☒ $x'' - 4x = -3, \quad x(0) = \frac{7}{4}, \quad x'(0) = -4$ ✓
- ☐ $x'' + 4x = -3, \quad x(0) = \frac{7}{4}, \quad x'(0) = 0$
- ☐ $x'' + 4x = 3, \quad x(0) = \frac{7}{4}, \quad x'(0) = 3$

Your answer is correct.

The correct answer is:

$$x'' - 4x = -3, \quad x(0) = \frac{7}{4}, \quad x'(0) = -4$$

Question 2

Incorrect

Mark 0.00 out of 1.00

Applying the Laplace transform approach to the system

$$\begin{cases} x' + y = 0, & x(0) = \frac{7}{4} \\ 4x + y' = 3, & y(0) = 4 \end{cases}$$

we obtain the following system, where $X(s) = \mathcal{L}\{x\}(s)$ and

$$Y(s) = \mathcal{L}\{y\}(s) :$$

Select one:

☐
$$\begin{cases} sX + Y = 0 \\ 4X + sY = 3 \end{cases}$$

☐
$$\begin{cases} sX + Y = 0 \\ 4sX + s^2Y = 3 \end{cases}$$

☒
$$\begin{cases} 4sX + 4Y = 7 \\ 4sX + s^2Y = 3 + 4s \end{cases} \quad \times$$

☐
$$\begin{cases} 4sX + 4Y = 7 \\ 4X + sY = 7 \end{cases}$$

Your answer is incorrect.

The correct answer is:
$$\begin{cases} 4sX + 4Y = 7 \\ 4X + sY = 7 \end{cases}$$

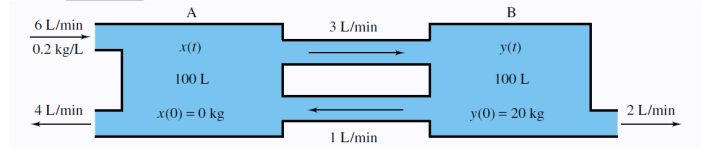
Question 3

Correct

Mark 1.00 out of 1.00

Two large tanks, each holding 100 L of liquid, are interconnected by pipes, with the liquid flowing from tank A into tank B at a rate of 3 L/min and from B into A at a rate of 1 L/min (see the figure below). The liquid inside each tank is kept well stirred. A brine solution with a concentration of 0.2 kg/L of salt flows into tank A at a rate of 6 L/min. The (diluted) solution flows out of the system from tank A at 4 L/min and from tank B at 2 L/min. Choose correct system of equations corresponding to this problem, if $x(t)$ is amount of salt in the tank

A and $y(t)$ is amount of salt in the tank B at the moment t .



Select one:

- ☐ $\begin{cases} x' = 1.2 - 0.01y + 0.07x \\ y' = 0.03y - 0.03x \end{cases}$
- ☒ $\begin{cases} x' = 1.2 + 0.01y - 0.07x \\ y' = 0.03x - 0.03y \end{cases}$ ✓
- ☐ $\begin{cases} x' = 1.2 + 0.03y - 0.01x \\ y' = 0.03y - 0.03x \end{cases}$
- ☐ $\begin{cases} x' = 0.03y - 0.07x \\ y' = 0.03x - 0.03y \end{cases}$

Your answer is correct.

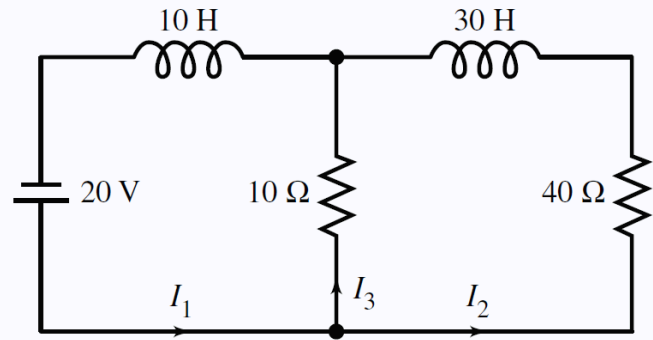
The correct answer is: $\begin{cases} x' = 1.2 + 0.01y - 0.07x \\ y' = 0.03x - 0.03y \end{cases}$

Question 4

Correct

Mark 1.00 out of 1.00

Find a system of differential equations for the currents in the network given in the schematic diagram below:



Select one:

☒
$$\begin{cases} I_1 = I_2 + I_3 \\ I_1' + I_3 = 2 \\ 3I_2' + 4I_2 - I_3 = 0 \end{cases}$$
 ✓

☐
$$\begin{cases} I_1 = I_2 + I_3 \\ I_1' - I_3 = 2 \\ 3I_2' - I_2 - I_3 = 0 \end{cases}$$

☐
$$\begin{cases} I_1 = I_2 + I_3 \\ I_1' - I_3 = 2 \\ 3I_2' + 4I_2 - I_3 = 0 \end{cases}$$

☐
$$\begin{cases} I_1 = I_2 + I_3 \\ I_1' + I_3 = 2 \\ 3I_2' - I_2 - I_3 = 0 \end{cases}$$

Your answer is correct.

The correct answer is:
$$\begin{cases} I_1 = I_2 + I_3 \\ I_1' + I_3 = 2 \\ 3I_2' + 4I_2 - I_3 = 0 \end{cases}$$

Question 5

Correct

Mark 1.00 out of 1.00

The normal form of the differential equation

$(1 - t^2)y'' - 2ty' + 2y = 0$ is $\mathbf{x}' = A\mathbf{x}$, where

$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $x_1 = y$, $x_2 = y'$, and the matrix A is

Select one:

☒ $\frac{1}{1-t^2} \begin{bmatrix} 0 & 1-t^2 \\ -2 & 2t \end{bmatrix}$ ✓

☐ $\frac{1}{1-t^2} \begin{bmatrix} 1-t^2 & 0 \\ 2t & -2 \end{bmatrix}$

☐ $\frac{1}{1-t^2} \begin{bmatrix} 0 & -2 \\ 1-t^2 & 2t \end{bmatrix}$

☐ $\frac{1}{1-t^2} \begin{bmatrix} -2 & 0 \\ 2t & 1-t^2 \end{bmatrix}$

Your answer is correct.

The correct answer is:

$\frac{1}{1-t^2} \begin{bmatrix} 0 & 1-t^2 \\ -2 & 2t \end{bmatrix}$